

## Experiment 9

**Aim:** To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.

### **Theory:**

Docker is a popular platform that enables developers to build, package, and deploy applications as lightweight, portable, and self-sufficient containers. These containers encapsulate all the necessary dependencies and libraries required for an application to run, ensuring consistency across different environments. Here is a theoretical overview of

### **Docker:**

#### **Containerization:**

Docker utilizes containerization technology to create isolated environments for applications. Containers are lightweight, standalone, and executable packages that include everything needed to run an application, such as code, runtime, system tools, libraries, and settings. This isolation ensures that applications run consistently across different environments, from development to production.

### **Docker Engine:**

At the core of Docker is the Docker Engine, which is responsible for building, running, and managing containers. It consists of the Docker daemon, which manages containers, images, networks, and volumes, and the Docker client, which allows users to interact with the daemon through the Docker API.

### **Docker Images:**

Docker images are read-only templates used to create containers. They contain the application code, runtime, libraries, dependencies, and other files needed to run the application. Images are built using Dockerfiles, which are text files that define the steps needed to create the image.

### **Docker Containers:**

Containers are instances of Docker images that are running as isolated processes on a host machine. They are lightweight, portable, and can be easily started, stopped, moved, and deleted. Containers provide a consistent environment for applications to run, regardless of the underlying infrastructure.

### **Benefits of Docker:**

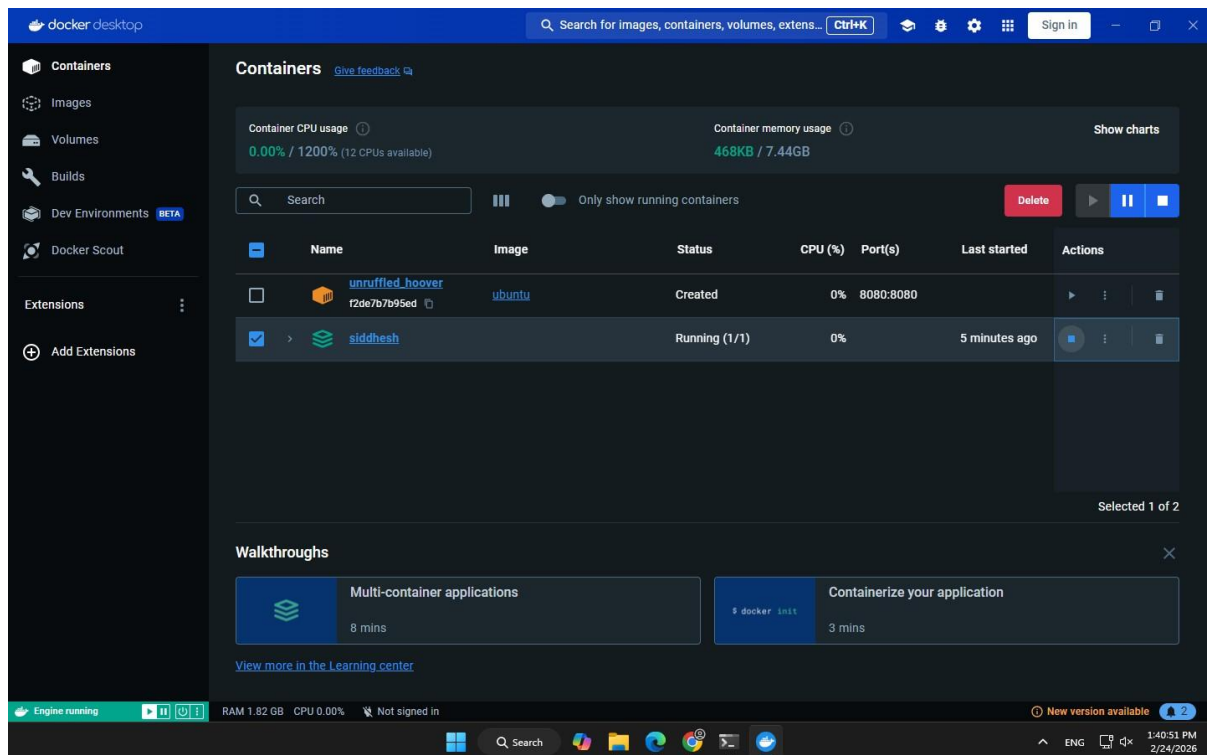
**Portability:** Docker containers can run on any platform that supports Docker, making it easy to deploy applications across different environments.

**Efficiency:** Containers share the host OS kernel, reducing overhead and improving resource utilization.

**Isolation:** Containers provide a level of isolation that helps prevent conflicts between applications and dependencies.

**Scalability:** Docker enables easy scaling of applications by quickly spinning up additional containers.

**Consistency:** Docker ensures that applications run the same way in development, testing, and production environments.



```
C:\Users\Lenovo>docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
run      Create and run a new container from an image
exec     Execute a command in a running container
ps       List containers
build    Build an image from a Dockerfile
pull     Download an image from a registry
push     Upload an image to a registry
images   List images
login    Log in to a registry
logout   Log out from a registry
search   Search Docker Hub for images
version  Show the Docker version information
info     Display system-wide information

Management Commands:
builder  Manage builds
buildx*  Docker Buildx (Docker Inc., v0.12.1-desktop.4)
compose* Docker Compose (Docker Inc., v2.24.6-desktop.1)
container Manage containers
context  Manage contexts
debug*   Get a shell into any image or container. (Docker Inc., 0.0.24)
dev*     Docker Dev Environments (Docker Inc., v0.1.0)
extension* Manages Docker extensions (Docker Inc., v0.2.22)
feedback* Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
image    Manage images
init*    Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
manifest Manage Docker image manifests and manifest lists
network  Manage networks
plugin   Manage plugins
sbom*    View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
scout*   Docker Scout (Docker Inc., v1.5.0)
system   Manage Docker
trust    Manage trust on Docker images
volume   Manage volumes
```

```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4debf41
```

```
C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker search ubuntu
```

| NAME                           | DESCRIPTION                                     | STARS | OFFICIAL |
|--------------------------------|-------------------------------------------------|-------|----------|
| ubuntu                         | Ubuntu is a Debian-based Linux operating sys... | 17789 | [OK]     |
| ubuntu/squid                   | Squid is a caching proxy for the Web. Long-t... | 124   |          |
| ubuntu/nginx                   | Nginx, a high-performance reverse proxy & we... | 139   |          |
| ubuntu/cortex                  | Cortex provides storage for Prometheus. Long... | 4     |          |
| ubuntu/bind9                   | BIND 9 is a very flexible, full-featured DNS... | 117   |          |
| ubuntu/kafka                   | Apache Kafka, a distributed event streaming ... | 60    |          |
| ubuntu/zookeeper               | ZooKeeper maintains configuration informatio... | 14    |          |
| ubuntu/apache2                 | Apache, a secure & extensible open-source HT... | 102   |          |
| ubuntu/prometheus              | Prometheus is a systems and service monitori... | 78    |          |
| ubuntu/mysql                   | MySQL open source fast, stable, multi-thread... | 72    |          |
| ubuntu/dotnet-aspnet           | Chiselled Ubuntu runtime image for ASP.NET a... | 26    |          |
| ubuntu/jre                     | Chiseled Java runtime based on Ubuntu. Long...  | 24    |          |
| ubuntu/postgres                | PostgreSQL is an open source object-relatio...  | 42    |          |
| ubuntu/python                  | A chiselled Ubuntu rock with the Python runt... | 31    |          |
| ubuntu/redis                   | Redis, an open source key-value store. Long...  | 24    |          |
| ubuntu/dotnet-runtime          | Chiselled Ubuntu runtime image for .NET apps... | 24    |          |
| ubuntu/dotnet-deps             | Chiselled Ubuntu for self-contained .NET & A... | 16    |          |
| ubuntu/grafana                 | Grafana, a feature rich metrics dashboard & ... | 12    |          |
| ubuntu/memcached               | Memcached, in-memory keyvalue store for smal... | 6     |          |
| ubuntu/prometheus-alertmanager | Alertmanager handles client alerts from Prom... | 10    |          |
| ubuntu/cassandra               | Cassandra, an open source NoSQL distributed ... | 3     |          |
| ubuntu/mlflow                  | MLFlow: for managing the machine learning li... | 6     |          |
| ubuntu/loki                    | Grafana Loki, a log aggregation system like ... | 2     |          |
| ubuntu/mimir                   | Ubuntu ROCK for Mimir, a horizontally scalab... | 0     |          |
| ubuntu/opentelemetry-collector | This is a rock for the OpenTelemetry Collect... | 0     |          |

```
C:\Users\Lenovo>docker search --filter=stars=1000 ubuntu
```

| NAME   | DESCRIPTION                                     | STARS | OFFICIAL |
|--------|-------------------------------------------------|-------|----------|
| ubuntu | Ubuntu is a Debian-based Linux operating sys... | 17789 | [OK]     |

```
C:\Users\Lenovo>docker search mysql
```

| NAME                | DESCRIPTION                                     | STARS | OFFICIAL |
|---------------------|-------------------------------------------------|-------|----------|
| mysql               | MySQL is a widely used, open-source relation... | 16070 | [OK]     |
| bitnami/mysql       | Bitnami Secure Image for mysql                  | 147   |          |
| circleci/mysql      | MySQL is a widely used, open-source relation... | 32    |          |
| bitnamicharts/mysql | Bitnami Helm chart for MySQL                    | 0     |          |
| cimg/mysql          |                                                 | 3     |          |
| ubuntu/mysql        | MySQL open source fast, stable, multi-thread... | 72    |          |
| google/mysql        | MySQL server for Google Compute Engine          | 26    |          |
| linuxserver/mysql   | A Mysql container, brought to you by LinuxSe... | 48    |          |
| elestio/mysql       | MySQL, verified and packaged by Elestio         | 3     |          |
| alpine/mysql        | mysql client                                    | 7     |          |
| docksal/mysql       | MySQL service images for Docksal - https://d... | 0     |          |
| eclipse/mysql       | MySQL 5.7, curl, rsync                          | 3     |          |
| ilios/mysql         | MySQL configured for running Ilios              | 1     |          |
| datajoint/mysql     | MySQL image pre-configured to work smoothly ... | 2     |          |
| ddev/mysql          | ARM64 base images for ddev-dbserver-mysql-8...  | 2     |          |
| mirantis/mysql      |                                                 | 0     |          |
| mysql/mysql-server  | Optimized MySQL Server Docker images. Create... | 1033  |          |
| corpusops/mysql     | https://github.com/corpusops/docker-images/     | 0     |          |
| cleanstart/mysql    | Secure by Design, Built for Speed, Hardened ... | 0     |          |
| vulhub/mysql        |                                                 | 1     |          |
| cbioportal/mysql    | This repository hosts MySQL database images ... | 1     |          |
| vitess/mysql        | Lightweight image to run MySQL with Vitess      | 1     |          |
| mysql/mysql-router  | MySQL Router provides transparent routing be... | 28    |          |
| mysql/mysql-cluster | Experimental MySQL Cluster Docker images. Cr... | 100   |          |
| nasqueron/mysql     |                                                 | 1     |          |

```
C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495efff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
1. Sign in to your Docker account → docker login
2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker images
```

| REPOSITORY                      | TAG                                     | IMAGE ID     | CREATED       | SIZE   |
|---------------------------------|-----------------------------------------|--------------|---------------|--------|
| ubuntu                          | latest                                  | bdbabce66f1b | 13 days ago   | 78.1MB |
| ubuntu                          | <none>                                  | 802541663949 | 6 months ago  | 78.1MB |
| ubuntu                          | <none>                                  | ca2b0f26964c | 24 months ago | 77.9MB |
| redis                           | <none>                                  | 170a1e90f843 | 2 years ago   | 138MB  |
| docker/dev-environments-default | stable-1                                | 7eb85f44c229 | 3 years ago   | 610MB  |
| crazymax/linguist               | 7.20.0                                  | dc5f9ae3be6a | 3 years ago   | 72.2MB |
| docker/desktop-git-helper       | 5a4fc126aadcd3f6cc3a011aa991de982ae7000 | efe2d67c403b | 4 years ago   | 44.2MB |

```
C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit

C:\Users\Lenovo>docker run -it ubuntu
root@2ddd2c7718ec:/# admin
bash: admin: command not found
root@2ddd2c7718ec:/# ^C
root@2ddd2c7718ec:/# exit
exit

C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb708ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f5f11d6b03f7bf18
Status: Downloaded newer image for httpd:latest
9f106d7c64c6c3703dc235fe26ca50a03a9876c261d5b29b20a30b006922393d

C:\Users\Lenovo>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
357926362d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
af5bbbf473e08d0b9a7264168926d9712c94a6d94497687fed89179d31e770ccd

C:\Users\Lenovo>docker run -d ubuntu
17ae4f0c3240c5b5ba81df3d23be8a1eef290d48684f89b47335567685514b7a

C:\Users\Lenovo>docker ps
```

| CONTAINER ID | IMAGE                                    | COMMAND            | CREATED            | STATUS            | PORTS  | NAMES             |
|--------------|------------------------------------------|--------------------|--------------------|-------------------|--------|-------------------|
| 9f10642c64c6 | httpd                                    | "httpd-foreground" | About a minute ago | Up About a minute | 80/tcp | wonderful_babbage |
| a388ed3e2ae7 | docker/dev-environments-default:stable-1 | "sleep infinity"   | 23 months ago      | Up 12 minutes     |        | siddhesh-app-1    |